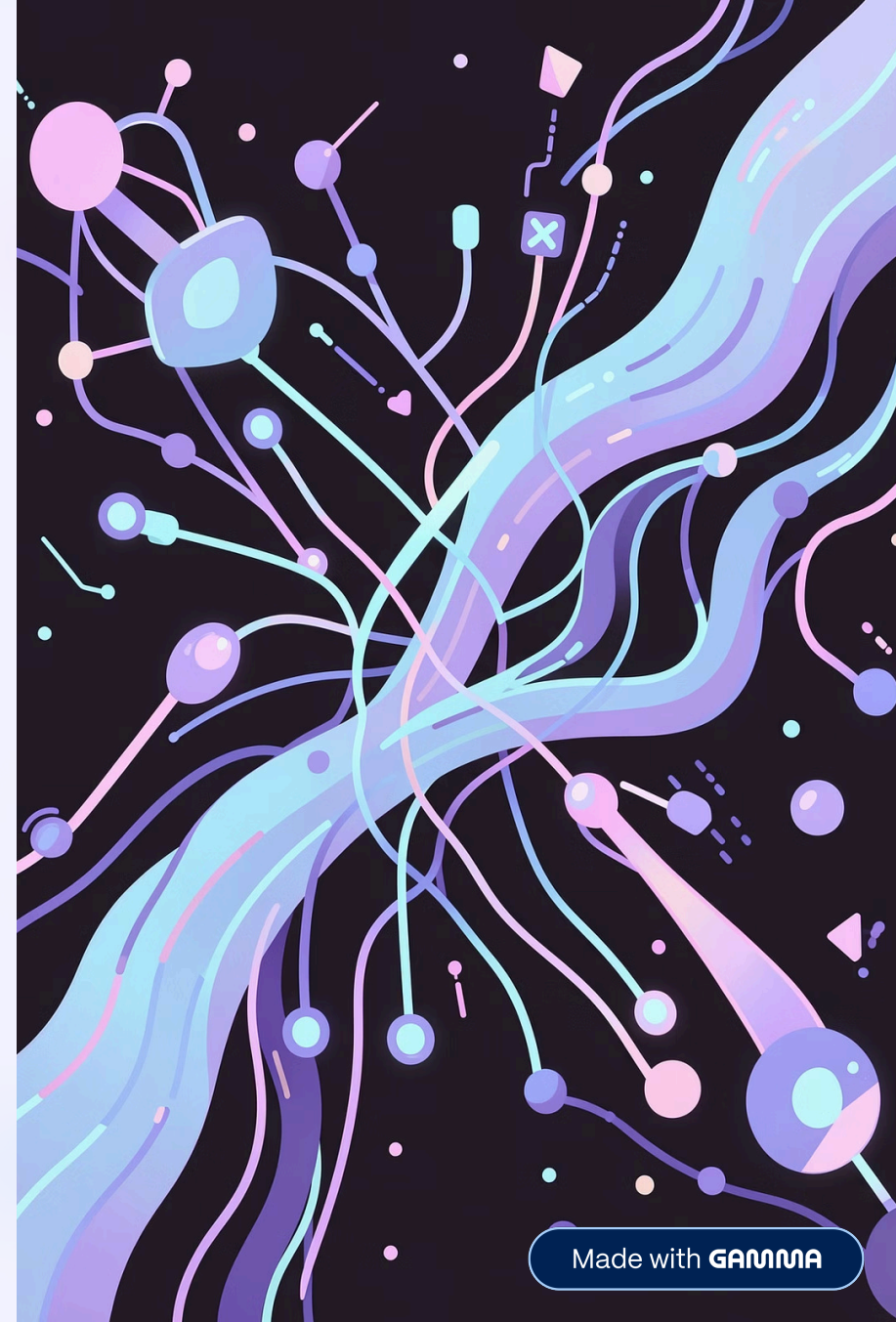
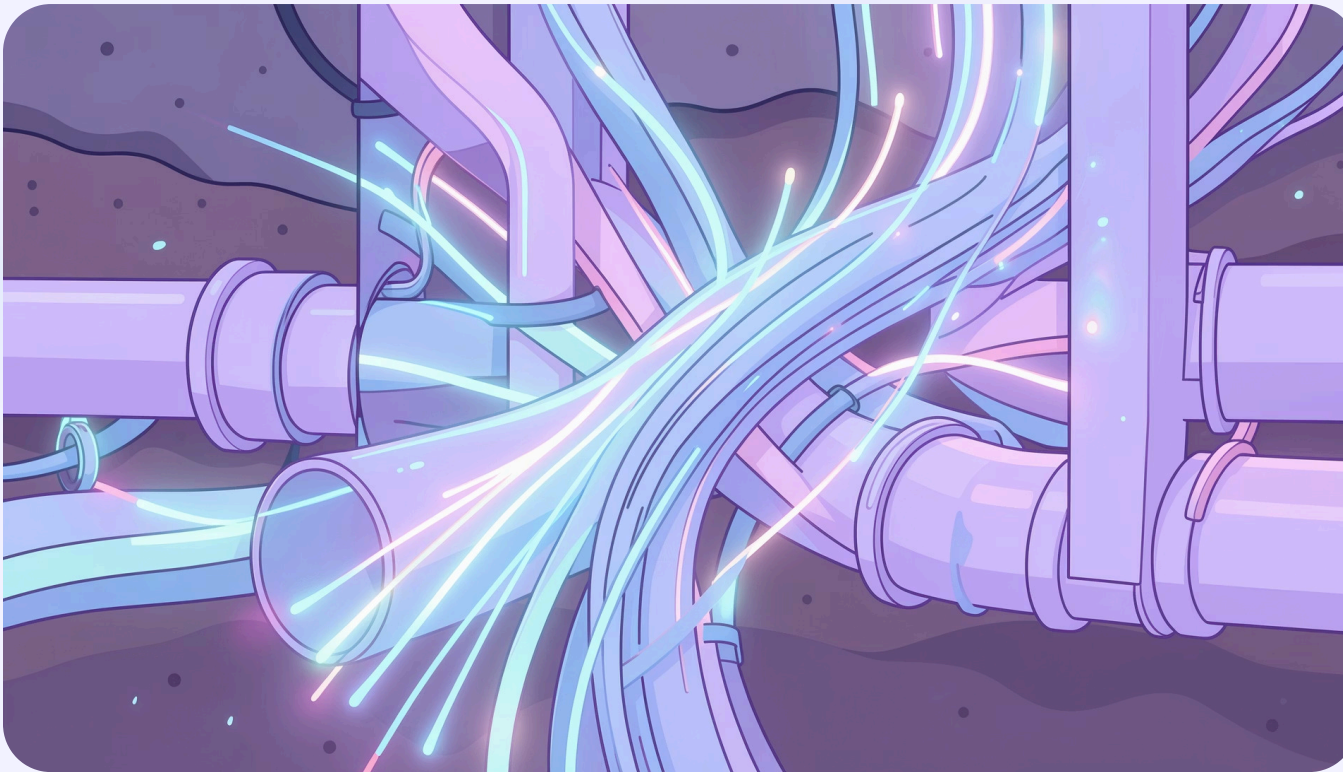


The Modern Web: From Foundations to Frontiers

A workshop journey through the technologies, tools, and ideas shaping how the web is built – and where it's headed next.



The Internet's Backbone — How It All Connects



More Than Websites

The internet is a global network of networks — a physical infrastructure of cables, routers, and protocols that powers everything from email to streaming.

From Electricity to the Web

Like electricity transformed industry, the internet has reshaped communication, commerce, and culture on a planetary scale.

Communication Protocols

Computers "talk" using standardized rules — TCP/IP, DNS, and HTTP — ensuring data travels reliably across the globe in milliseconds.

Building Blocks of the Web

Every website you visit is built on three core technologies. Master them, and you can build anything.



HTML

The skeleton of the web. Defines structure, content, and meaning using semantic tags.



CSS

The style layer. Controls layout, color, typography, and responsive behavior across devices.



JavaScript

The brain. Adds interactivity, logic, and dynamic behavior – turning static pages into living apps.

 **From "Never Seen Code" to "I Built This Site"** – a hands-on crash course takes you from zero to a deployed web page in hours.

Beyond the Basics — Advanced Web Technologies

→ Progressive Web Apps (PWAs)

Offline mode, push notifications, and home screen installation — all from a browser.

→ Vanilla Web Components

Native, fast, framework-free reusable UI elements built with standard JavaScript APIs.

→ HTTP/3 & Service Workers

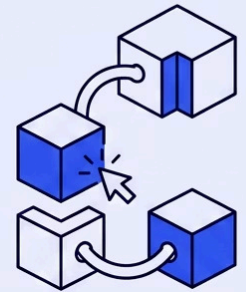
Next-gen protocols and caching strategies that make apps faster and more resilient.

→ WebAssembly & Jamstack

Run near-native code in the browser and deploy blazing-fast static architectures.



PWAs
Installable, offline apps.



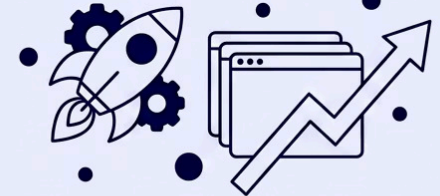
WEB COMPONENTS
Native reusable elements.

HTTP/3 + WORKERS

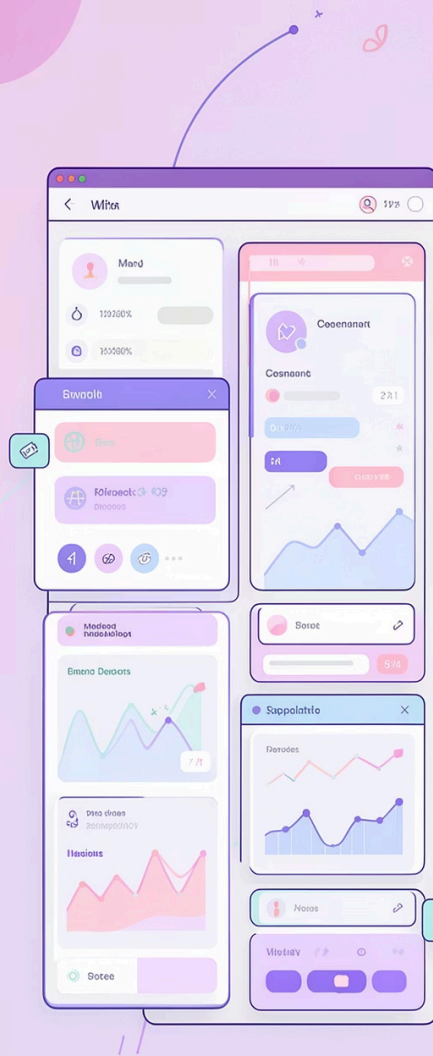


Speed and resilience.

WASM + JAMSTACK



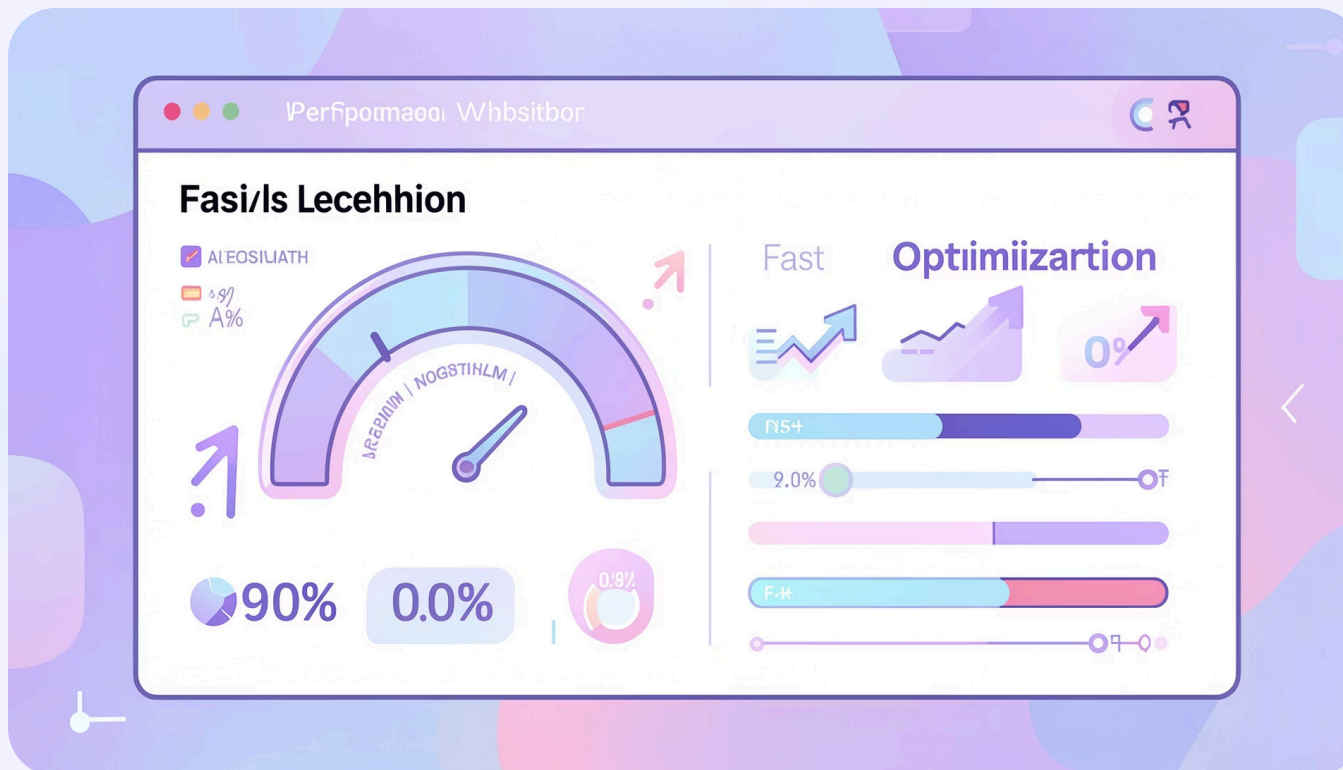
Performance, modern arch.



Evolution of the Web

From static text pages to immersive, app-like experiences – the web has transformed beyond recognition. Understanding this evolution helps us build better for what comes next.

Performance & User Experience — The Crucial Edge



Why Speed Is a Feature

Users abandon pages that take more than 3 seconds to load. Performance is not optional – it's a core product requirement.

Auditing

Use Lighthouse and Chrome DevTools to find bottlenecks.

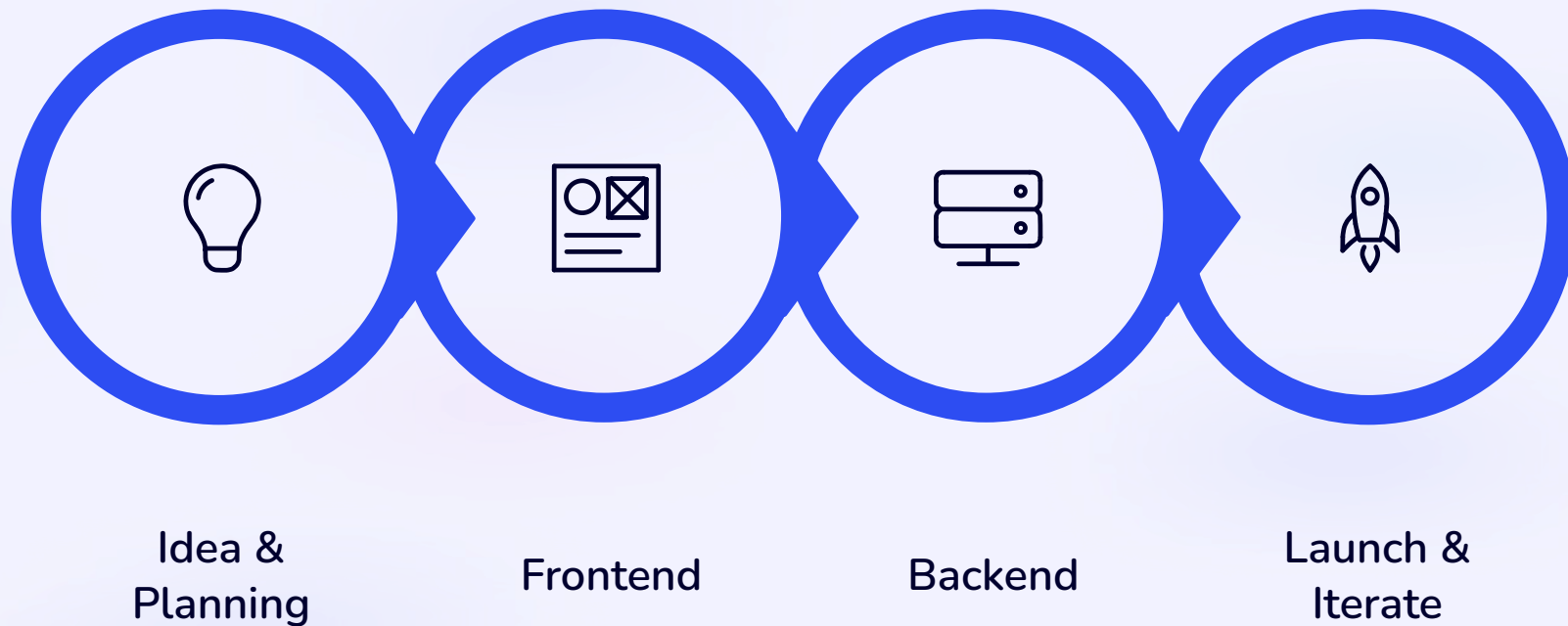
Image & Font Optimization

Modern formats (WebP, AVIF) and lazy loading cut load times dramatically.

CSS Management

Critical CSS inlining and tree-shaking reduce render-blocking overhead.

Full Stack Foundations — Building Complete Applications



Modern web applications require fluency across the entire stack. From routing and data loading to CSS architecture and user interactions — every layer matters for building robust, production-ready apps.

The Future Is Now — AI & Evolving Web Engineering

AI-Assisted Development

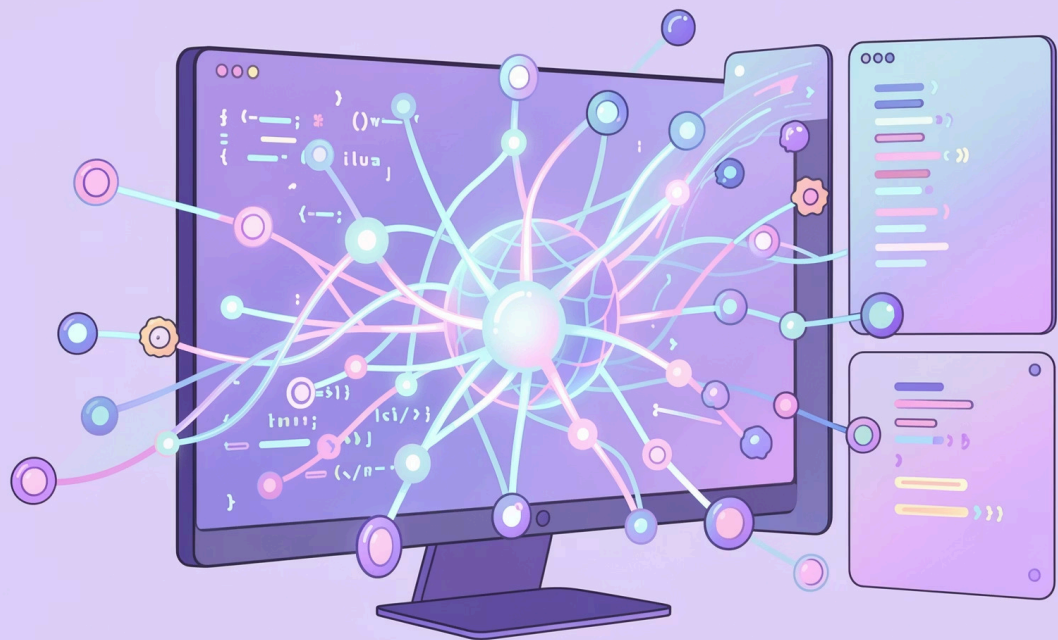
AI tools are changing how we write, debug, and design code — accelerating development and lowering the barrier to entry.

Build with Node.js

Node.js lets you use JavaScript on the server — enabling you to build full websites and APIs with one language.

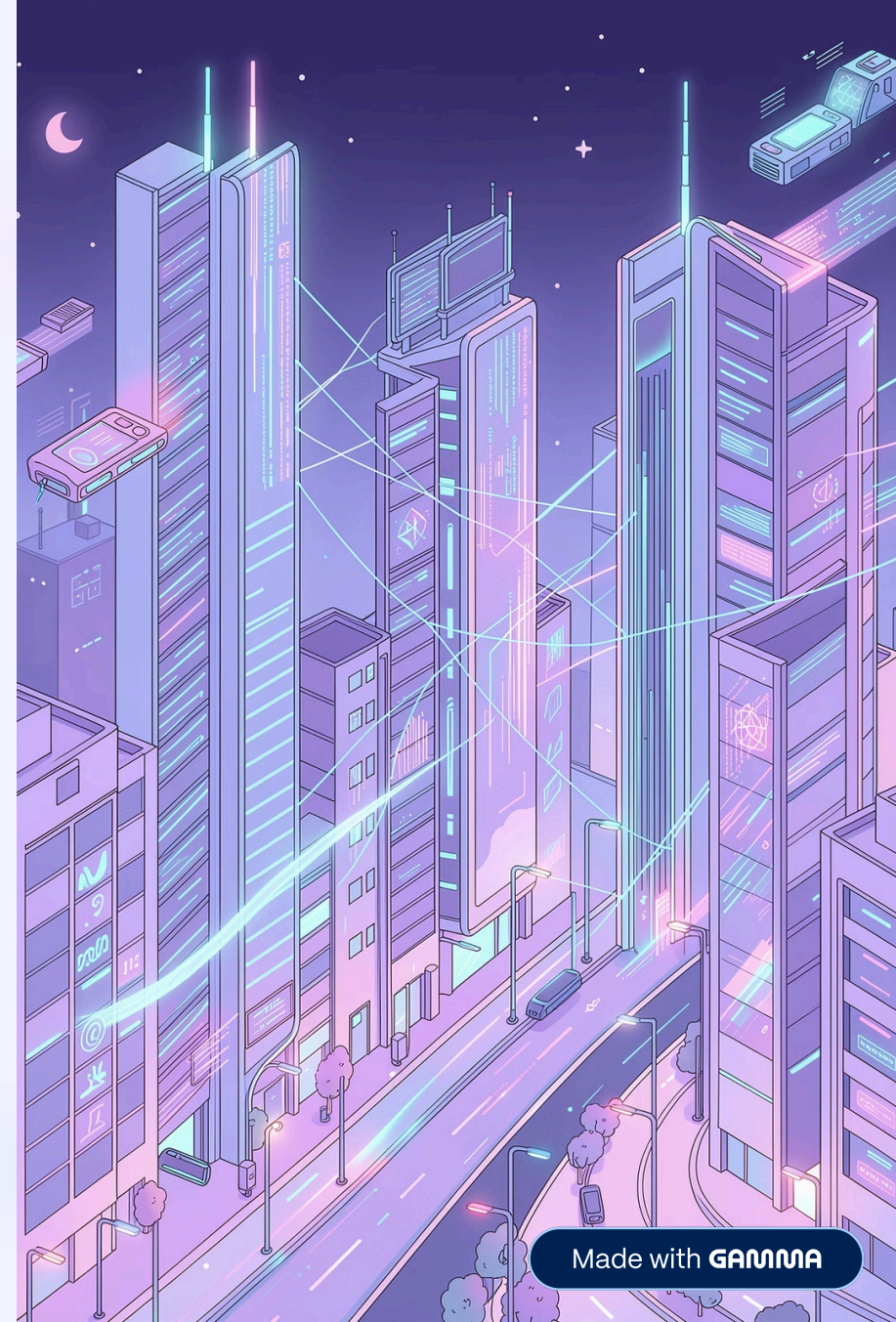
Stay Relevant

The web evolves fast. Continuous learning through workshops, communities, and documentation is your greatest competitive advantage.



Innovate. Build. Connect.

The web is the most powerful creative platform in human history. The tools are in your hands — now it's time to use them.



Made with **GAMMA**

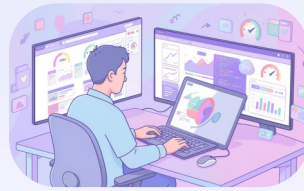
YOUR NEXT STEPS

Embrace the Modern Web



Foundational Skills

Master HTML, CSS, and JavaScript before reaching for frameworks. Strong fundamentals unlock everything else.



Advanced Techniques

Explore PWAs, Web Components, and modern protocols to push what your apps can do.



Keep Learning

Join communities, read documentation, attend workshops. The web never stops evolving – and neither should you.

✅ **Start today:** Pick one idea from this workshop and build it. Your next great web project begins with a single line of code.